Facilities and Other Resources

ONR FY2027 YIP, NOFO N0001426SF004. PI: Sterling G. Baird, Brigham Young University

PI laboratory, BYU Department of Mechanical Engineering. A powder metallurgy laboratory [TODO Sterling: square footage and room] holds an AMAZEMET rePOWDER ultrasonic atomizer with an induction crucible module, which makes alloy powder from elemental and master-alloy feedstock weighed into machined cups, about four compositions per week. For this project the cups are oxygen-free copper and the powders are copper alloys, melted at up to 1,300 °C in graphite crucibles and atomized on coated carbon-carbon plates. For melts that contain nickel or iron, the crucible is sprayed with boron nitride and the nozzle is boron nitride [TODO Sterling: confirm the copper charge size]. Sieves, a Hall flowmeter funnel, and dry powder storage support the powder checks. The room has a canopy hood, a fume hood, an industrial dehumidifier, and electrical, gas, compressed air, and chilled water services, installed with BYU Environmental Health and Safety. The laboratory connects directly to the BYU Microscopy Center by a door with no public access.

BYU Microscopy Center. Scanning electron microscopes with energy-dispersive X-ray spectroscopy (EDS) and electron backscatter diffraction (EBSD) (Apreo C, Helios 600 with a focused ion beam, and Verios), and a Tecnai TF-20 transmission electron microscope. Instruments are shared and have high availability.

Other BYU facilities. Metallographic preparation (PX500 automatic and PX300 manual polishers, vibratory polishers, fume hood for etching, and an electropolisher); a Keyence VHX-7000 digital optical microscope; the BYU X-ray Diffraction Facility; the BYU Spectroscopic Facility (Agilent 8900 ICP-MS/MS for composition, and XPS); and the BYU Prototyping Lab (CNC machining for charge cups and test specimens). Tensile tests use [TODO Sterling: name the BYU load frame or test service]. Particle size distributions use [TODO Sterling: name the laser diffraction analyzer or service].

Computing. The BYU Office of Research Computing operates a shared cluster of about 28,000 CPU cores and 367 GPUs on 616 nodes (`https://rc.byu.edu`) [TODO Sterling: update the counts if they have changed since April 2026], available at no direct cost through fair-share scheduling. CALPHAD calculations use Thermo-Calc [TODO Sterling: confirm the BYU license includes a copper alloy database] and pycalphad.

Directed energy deposition. Single-stream, blown-powder laser DED sessions are bought on a purchase order from [TODO Sterling: facility], on its FormAlloy X2, at its external hourly rates. The X2 takes 50 to 200 μm powder, about 2 kg of each, and also melts powder pre-placed in grooves [TODO Sterling: laser, shielding].

Corrosion testing. The electrochemical screen runs on the PI laboratory's liquid-handling gantry (CubXL, Ursa Laboratories: a pipette, a vial capper, and two cameras on a scripted gantry with about 390 $\times$ 230 $\times$ 120 mm of travel), which positions small O-ring cells on polished specimens, with the potentiostat in Field 11. Immersion tests in substitute ocean water run in capped vials on the gantry's deck, which samples the solutions for ICP-MS at the BYU Spectroscopic Facility. Oxygen and carbon analysis of powder lots is bought from [TODO Sterling: service laboratory: inert-gas fusion for oxygen, combustion for carbon].

Government-owned facilities. None are required [TODO Sterling: revise if a Navy laboratory agrees to run flowing-seawater tests on final candidates].